

David Cain | <https://davidcain.uk>

|  | Hull, England | <https://www.linkedin.com/in/david-cain-811a81259>

Computer Science - First-Class Honours. 4-5 years of experience with C#. Microsoft Azure experience. 1 year of industry experience as test automation engineer at embedded electronics design company. Strong skills in C#, Web Development, Parallel Programming, Robot Framework, Python, C, Embedded Systems, Full Stack Technologies, Machine Learning.

Education:

BSc Computer Science with Industrial Experience – University of Hull

September 2021 – July 2025

- First-Class Honours
- Distributed Systems, Parallel Programming, Machine Learning, UI/UX, Placement Year

Year In Industry:

Automation Engineer & more – (Year in Industry) – Sabre Technology

July 2023 – July 2024

- Automated large-scale embedded system testing using Robot Framework to reduce manual test effort for team
- Collaborated across firmware/test teams in Git (tickets, pull requests, branching), and management in weekly meetings
- Developed internal tools to help team productivity:
 - Python LoRaWAN packet decoder library using YAML message schemas for extensibility
 - WPF frontend for a barcoding tool
 - Google Sheets Hex Message encoder and decoder
 - Documentation for tools and processes
 - Pick and Place file preprocessing script

Selected Projects:

- **Cryptography & Authentication API – Uni Assignment – (C#, ASP.NET Web API, Entity Framework)**
 - REST endpoints for SHA hashing and signing, RSA and AES encryption, accounts, API keys, and more
 - Implemented in ASP.NET Web API, and Entity Framework with C# client using calling the API for local services
- **OV7670 Camera API & Web Interface – Own Project – (Python, Flask, C, CMake, JS, HTML, CSS, KiCad)**
 - C RP2040 firmware and custom high frequency PCB to stream image data from bare-bones CMOS sensor module to Python Flask server over serial under microcontroller memory and clock limitations.
 - JS frontend uses the API to take images and set registers, rendering with HTML Canvas
- **Honours Stage Project – Uni Module – (JS, HTML, CSS, HTML Canvas)**
 - Created a zero-footprint web application for creating visual radiology reports for 3D medical scans.
 - Created a JavaScript rendering library for multiplanar reconstruction, slicing, segmentation, and rendering with a Web App interface and conducted user testing and evaluation
- **Parallel Programming – Uni Module – (CUDA, RUST, OpenGL)**
 - Utilised Cuda and multithreaded Rust to parallelise ray casting, bird flocking simulation, matrix multiplication, and more with massive performance boosts using memory tiling, locks, and more
- **Machine Learning – Uni Module – (TensorFlow, PyTorch, GAN, CNN)**
 - Created, trained, fine-tuned, and regularised deep CNNs and GANs in TensorFlow and PyTorch to solve written digit labelling and synthetic data generation. Accelerated with Google Colab cloud TPUs and GPUs
- **Full Portfolio:** <https://davidcain.uk>

Skills:

- C# • HTML, CSS, JavaScript • Entity Framework • SQL • Robot Framework • Python • Flask • Rust • Cuda
- Git • Microsoft Azure • C, C++ & CMake • Cuda • Rust • Postman
- Parallel Programming • AI, TensorFlow & PyTorch • Unity • Google Colab • Google Sheets + Excel
- Agile • Waterfall • UML • WordPress • Embedded Systems • Electronics • Art & Design

Achievements:

- **Best Certificate Academic Performance in Computer Science (University of Hull)** 2021 – 2022
- **AZ-900 Azure Cloud Fundamentals (Microsoft)** May 2022

Additional Work Experience:

Email Marketing & Lead Acquisition – Perimeter Site Security

October 2025 – Present

- Email lead acquisition and outreach
- Created web scraper scripts to gather contacts from online directories and websites

B2C Inbound Telesales Advisor – Clearanswer

July 2025 – October 2025

- Sales agent for BT and EE broadband, sims, mobiles, and TV
- Upsell products to inbound callers, explain technologies and specs to consumers, solve queries and complaints
- Use EE and BT SaaS apps for handling accounts and contracts
- Collaborate cross-department for non-sales needs
- Developed communication skills and deeper insight into network and broadband infrastructure

Student Demonstrator – University of Hull

September 2022 – July 2023

- Helped teach C# Object Oriented Programming to 1st year students in labs

Part-Time Work – Various

2019, 2020, 2021

- Work pre-university - various cafes, restaurants and fast paced environments